Dataset Description – VinBigData Chest X-ray Abnormalities Detection

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Dataset Overview

The VinBigData Chest X-ray Abnormalities Detection dataset is a large-scale medical imaging dataset released by Vingroup Big Data Institute (VinBigData) for research in automated chest X-ray analysis. It is hosted on Kaggle and contains DICOM-format chest radiographs annotated by experienced radiologists across 14 distinct pathological findings.

The dataset is designed for object detection tasks — each image may contain one or more labeled bounding boxes identifying the location and type of abnormality. The goal is to train a model that can localize and classify radiological findings in unseen chest X-rays, which has direct clinical value for assisting radiologists in high-volume screening environments.

Dataset Statistics

Attribute	Details
Source	VinBigData — Kaggle Competition
Image Format	DICOM (.dicom) → Converted to PNG
Total Training Images	15,000 chest X-rays
Total Test Images	3,000 chest X-rays
Image Resolution	Variable (up to 3000×4000 px original)
Preprocessed Resolution	512×512 px
Annotation Type	Bounding boxes (YOLO format)
Number of Classes	14 findings + "No finding"
Annotators per Image	3 independent radiologists
Images used (this project)	50 (development/testing phase)

Folder Structure

The Kaggle competition dataset is organized as follows:

Folder / File	Description
train/	Training DICOM images (15,000 files)
test/	Test DICOM images (3,000 files)
train.csv	Bounding box annotations for training images
sample_submission.csv	Submission format for Kaggle

train.csv — Column Descriptions

The annotation file train.csv contains one row per radiologist per finding per image. Each column is described below:

image_id *Type: Categorical* Unique identifier for each chest X-ray image. Corresponds directly to the DICOM filename (without extension). Used as the primary key to link images with their annotations.

class_name *Type: Categorical* Name of the radiological finding annotated by the radiologist. There are 14 pathology classes plus a special 'No finding' class indicating a normal scan with no abnormality.

class_id *Type: Numerical (Integer)* Numerical encoding of the class_name. Values range from 0 to 14, where 14 corresponds to 'No finding'. Used directly as the class label in YOLO annotation format.

rad_id *Type: Categorical* Identifier for the radiologist who provided the annotation. Each image is independently labeled by 3 radiologists (e.g., R1, R2, R3), enabling inter-annotator agreement analysis.

x_min *Type: Numerical (Float)* X-coordinate of the top-left corner of the bounding box in the original image's pixel space. Must be scaled to the preprocessed image resolution before use in training.

y_min *Type: Numerical (Float)* Y-coordinate of the top-left corner of the bounding box in the original image's pixel space. Must be scaled proportionally when the image is resized during preprocessing.

x_max *Type: Numerical (Float)* X-coordinate of the bottom-right corner of the bounding box. Together with x_min, defines the horizontal extent of the annotated region.

y_max *Type: Numerical (Float)* Y-coordinate of the bottom-right corner of the bounding box. Together with y_min, defines the vertical extent of the annotated region.

The 14 Pathological Finding Classes

The dataset covers 14 distinct radiological abnormalities. Each class corresponds to a specific finding identifiable on a chest X-ray. A 15th label — 'No finding' — is used for normal scans and carries no bounding box to annotate.

Class ID	Class Name	Description
0	Aortic enlargement	Widening of the aortic silhouette on CXR
1	Atelectasis	Partial or complete lung collapse
2	Calcification	Calcium deposits visible in lung tissue or vessels
3	Cardiomegaly	Enlarged heart shadow beyond normal cardiac ratio
4	Consolidation	Fluid/material filling alveolar air spaces

Class ID	Class Name	Description
5	ILD	Interstitial lung disease — diffuse lung tissue changes
6	Infiltration	Hazy opacities suggesting infection or inflammation
7	Lung Opacity	General areas of increased density in the lung field
8	Nodule/Mass	Discrete round lesions within the lung parenchyma
9	Other lesion	Miscellaneous findings not covered by other classes
10	Pleural effusion	Fluid accumulation in the pleural space
11	Pleural thickening	Thickening of the pleural lining
12	Pneumothorax	Air in the pleural cavity causing lung collapse
13	Pulmonary fibrosis	Scarring and stiffening of lung tissue
14	No finding	Normal scan — no bounding box, empty label file

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Annotation Format (YOLO)

Annotations are converted from the raw CSV bounding box format to YOLO format for use in object detection model training. Each image has a corresponding .txt label file with one line per bounding box.

Format: class_id center_x center_y width height

All values except class_id are normalized to the range [0, 1] relative to the image dimensions. Images with 'No finding' have an empty .txt file. Bounding box coordinates from the original DICOM resolution are scaled by a factor of 0.5 to match the preprocessed 512×512 resolution before normalization.

Ethical and Practical Considerations

- Only a **small subset of images** was used for demonstration.
- The dataset contains **no personally identifiable patient information**.
- Usage is strictly for **academic and research purposes**.

Project Summary

This dataset provides a clinically relevant and technically rich foundation for exploring medical image preprocessing and abnormality detection workflows. By converting DICOM images into PNG and applying systematic preprocessing techniques, the dataset becomes clean, consistent, and suitable for developing AI models aimed at assisting automated lung abnormality detection.